

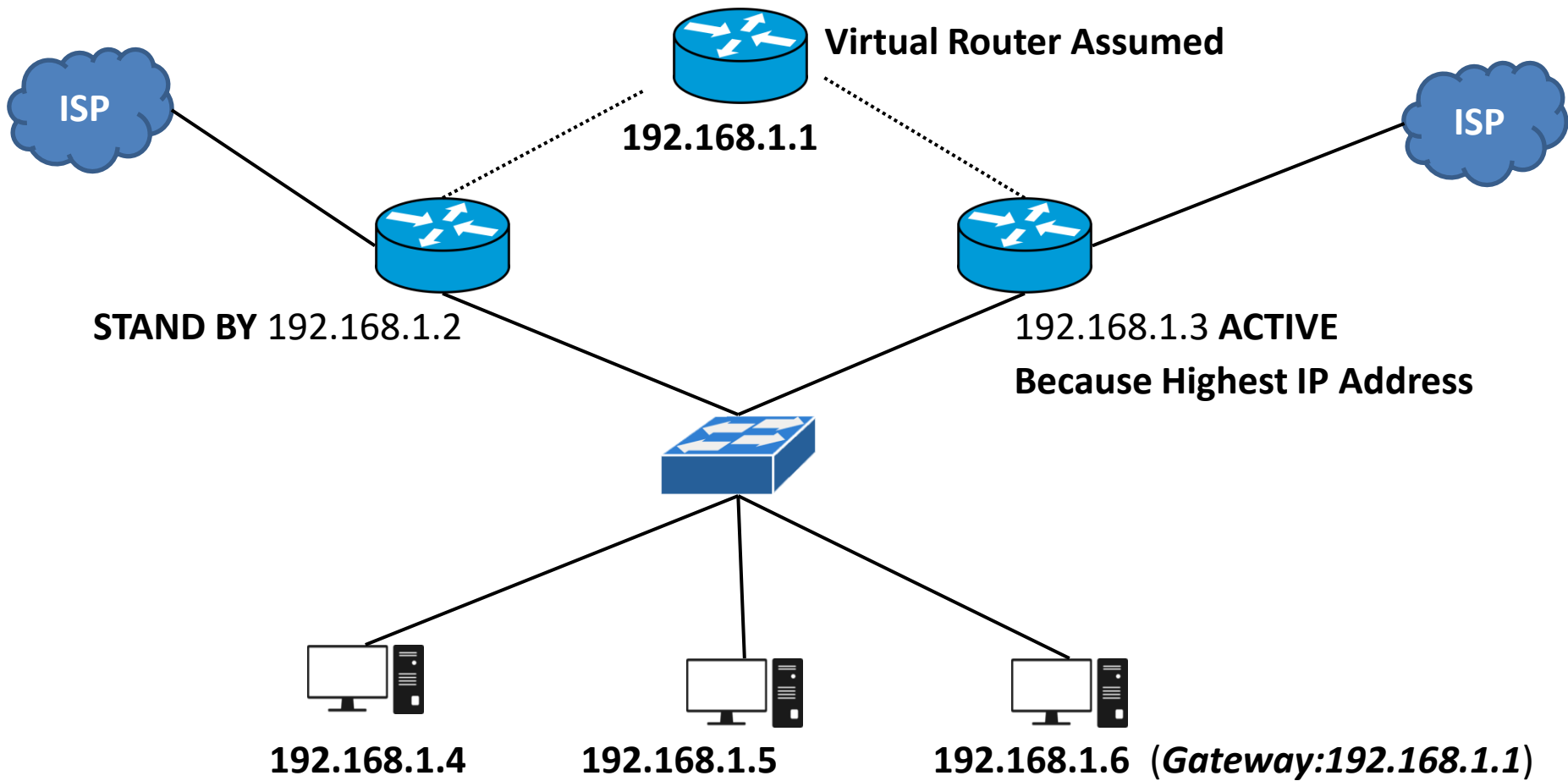
HSRP (Hot Standby Router Protocol):

HSRP (Hot Standby Router Protocol) is a Cisco proprietary redundancy protocol for establishing a fault-tolerant default gateway. HSRP enables a group of routers to work together to present the appearance of a single virtual router to the hosts on a LAN. This virtual router is assigned a virtual IP address that serves as the default gateway for devices in the network.

Purpose of HSRP

HSRP is designed to provide high network availability by providing redundancy for IP routing. If the active router fails, the standby router takes over the IP address and continues to provide routing services without disruption. This process is transparent to end devices, ensuring uninterrupted network service.

HSRP : Hot Standby Router Protocol:



Key Purposes of HSRP:

Virtual Router:

- HSRP creates a virtual router represented by a virtual IP address and a virtual MAC address. Hosts on the LAN configure this virtual IP as their default gateway.
- The virtual router has its own IP address (virtual IP) and MAC address (virtual MAC) that are shared by two or more physical routers.

Active and Standby Routers:

- **Active Router:** The router currently forwarding packets sent to the virtual IP address. This router actively routes traffic for the hosts.
- **Standby Router:** The router that monitors the status of the active router and takes over if the active router fails.

HSRP States:

Initial: The starting state when HSRP is first enabled.

Learn: The router is waiting to hear from the active router to learn the virtual IP address.

Listen: The router knows the virtual IP address but is not the active or standby router.

Speak: The router is sending periodic HSRP hello messages and is actively participating in the election of the active or standby router.

Standby: The router is a candidate to become the next active router and is listening for periodic hello messages from the active router.

Active: The router is currently forwarding packets that are sent to the group's virtual IP address.

HSRP Priority:

- Priority determines the active router in the group. The router with the highest priority value becomes the active router.
- If two routers have the same priority, the router with the highest IP address becomes the active router.
- The default priority for **HSRP (Hot Standby Router Protocol)** is **100**.

HSRP Configuration: -

```
Router(Config)#interface GigabitEthernet0/1
```

```
Router(Config)# ip address 192.168.1.2 255.255.255.0
```

```
Router(Config)# standby 1 ip 192.168.1.1 (virtual Router)
```

```
Router(Config)# standby 1 priority 110
```

```
Router(Config)# standby 1 preempt
```

Explanation:

standby 1 ip 192.168.1.1: Configures the virtual IP address for HSRP group 1.

standby 1 priority 110: Sets the router's HSRP priority to **110**.

standby 1 preempt: Enables preemption, allowing the router to take over as the active router if its priority is higher than active router.